

# Reproducing the Robustness Evaluation of Defensive Distillation under the Carlini–Wagner $L_2$ Attack

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**Abstract**—This report presents a reproduction study of the core experimental pipeline in *Towards Evaluating the Robustness of Neural Networks*, with a particular focus on defensive distillation and the targeted Carlini–Wagner (C&W)  $L_2$  attack. We reproduce three main components of the original work: undistilled baseline training, defensive distillation, and subsequent white-box adversarial evaluation under a strong optimization-based attack. The main reproduction is conducted on MNIST and CIFAR-10, and we further extend the study with a temperature sensitivity analysis of defensive distillation and an ImageNet-scale best/worst-target evaluation.

The reproduced results recover the principal qualitative conclusion of the original paper. Although defensive distillation increases the average perturbation magnitude required for successful attacks in some settings, it does not provide meaningful robustness against the targeted C&W  $L_2$  attack. In particular, attack success rates remain high for both distilled and undistilled models, indicating that the apparent defensive benefit of distillation should not be interpreted as genuine adversarial security.

At the quantitative level, our reproduction is more faithful in trend than in exact numerical magnitude. The strongest agreement with the original study is observed in attack success rates and in the overall vulnerability pattern under adaptive evaluation, while some discrepancies remain in the exact  $L_2$  distortion values and in the clean performance of the CIFAR-10 distilled model. We further analyze the implementation and optimization factors that affected reproducibility, including temperature scaling, knowledge-distillation loss design, engineering safeguards for stable attack evaluation, and the role of target-selection difficulty at larger scales. These extended analyses further reinforce the conclusion that strong adaptive attacks remain effective despite apparent defensive gains. Overall, our study supports the central argument of the original paper and reinforces the importance of strong adaptive attacks in robustness evaluation.

**Index Terms**—adversarial examples, robustness evaluation, defensive distillation, Carlini–Wagner attack, reproduction study, neural networks

## I. INTRODUCTION

Deep neural networks have achieved remarkable success across a wide range of machine learning tasks, including image classification, speech recognition, and natural language processing. Their strong empirical performance has led to rapid adoption in increasingly high-stakes settings, such as autonomous systems, security screening, and decision-support pipelines. However, the practical deployment of such models has been persistently challenged by the phenomenon of *adversarial examples*, namely inputs perturbed in a carefully

optimized yet often imperceptible manner so as to induce erroneous predictions [1], [2]. The existence of adversarial examples raises a fundamental question for modern machine learning: whether highly accurate neural networks can also be made reliably robust.

In response to this challenge, a large body of work has sought to improve adversarial robustness through both attack and defense mechanisms. Among early defense strategies, *defensive distillation* emerged as a particularly influential proposal [3]. By training a model on softened output distributions produced at elevated softmax temperature, defensive distillation was reported to substantially reduce the effectiveness of then-existing attack methods. Its appeal lay not only in its empirical promise, but also in its apparent simplicity: the defense could be applied to standard feed-forward neural networks with limited modification to the original training pipeline. At the time, this made defensive distillation one of the most prominent and widely discussed defenses against adversarial perturbations [3].

Nevertheless, the security value of any proposed defense depends critically on the strength of the attack used to evaluate it. A defense may appear effective simply because it invalidates a particular weak or poorly adapted attack, rather than because it genuinely increases the model’s resistance to adversarial manipulation. This distinction is central to the paper *Towards Evaluating the Robustness of Neural Networks* by Carlini and Wagner [4], which argues that robustness claims must be assessed against strong, optimization-based, and defense-aware adversaries. This perspective is also consistent with later work that formalized robustness evaluation through stronger first-order adversaries and robust optimization viewpoints [5].

The significance of the paper extends beyond its empirical findings on a single defense. More broadly, it reshaped the methodology of adversarial robustness evaluation by emphasizing that the failure of prior attacks should not be conflated with actual security. Instead, robustness must be interpreted as an upper-bounded property, to be stress-tested by attacks that are specifically designed to expose the weakest points of a defended model. For this reason, the Carlini–Wagner (C&W) attacks introduced in the paper have become standard baselines in subsequent research on adversarial machine learning, and later studies on obfuscated or masked gradients further reinforced this methodological lesson [4], [6].

In this report, we reproduce the core experimental logic of Carlini and Wagner’s study, with particular emphasis on the claim that defensive distillation does not confer genuine

robustness when confronted with a sufficiently strong targeted attack [4]. Concretely, we reproduce the undistilled baseline training pipeline, implement defensive distillation [3], and evaluate both distilled and undistilled models under the targeted C&W  $L_2$  attack [4]. Our goal is not merely to rerun code, but to examine whether the paper’s central conclusion can be recovered under an independently implemented experimental pipeline, and to identify the implementation choices that most strongly affect reproducibility.

Our reproduction study is guided by the following questions. First, can the clean performance of the undistilled and distilled models be reproduced with reasonable fidelity? Second, can the targeted C&W  $L_2$  attack achieve high success rates against defensively distilled networks, as reported in the original paper [4]? Third, to what extent do our reproduced numerical results align with the trends and conclusions presented by Carlini and Wagner? Finally, when discrepancies arise, what experimental or implementation factors most plausibly explain them?

By addressing these questions, this report aims to make two contributions. At the empirical level, it provides a structured reproduction of one of the most influential robustness evaluations in the adversarial machine learning literature [4]. At the methodological level, it highlights a broader lesson that remains highly relevant today: robust evaluation is inseparable from strong attack design, transparent experimental protocols, and careful interpretation of apparent defensive gains [5], [6].

## II. OVERVIEW OF THE ORIGINAL PAPER

### A. Problem Setting and Motivation

The original paper by Carlini and Wagner [4] is situated at a pivotal moment in the development of adversarial machine learning. By 2016–2017, it had already become well established that neural networks were vulnerable to adversarial perturbations [1], [2]. At the same time, several defense proposals had begun to appear, many of which claimed substantial robustness improvements under the attack algorithms then available. Defensive distillation [3] was among the most prominent of these approaches, having been reported to reduce attack success rates dramatically. This created an important methodological question: does such a defense truly improve robustness, or does it merely interfere with the specific attacks used in prior evaluations?

Carlini and Wagner address this question by reframing robustness evaluation as the search for stronger upper bounds on model vulnerability [4]. Rather than attempting to prove that a model is robust in a formal sense, they ask whether a sufficiently capable adversary can still construct successful adversarial examples. Under this perspective, a defense should not be considered secure simply because previously published attacks fail against it. Instead, security claims must survive adaptive attacks that are explicitly designed for the defended model.

### B. Core Idea of the Paper

The central idea of the paper is that defensive distillation does not fundamentally eliminate adversarial examples; rather, it causes certain earlier attack methods to become ineffective

or numerically unstable [4]. To demonstrate this point, the authors introduce a new family of attack algorithms tailored to three commonly used perturbation metrics, namely  $L_0$ ,  $L_2$ , and  $L_\infty$  [4]. These attacks are formulated as direct optimization problems and are designed to produce low-distortion targeted adversarial examples with substantially higher reliability than prior methods.

Among the proposed attacks, the  $L_2$ -based Carlini–Wagner attack is especially important because it became one of the most influential optimization-based attacks in the literature [4]. Its formulation balances two competing objectives: minimizing perturbation magnitude while ensuring that the perturbed input is classified as a chosen target label. Through an appropriate change of variables and a carefully designed objective function defined on logits rather than probabilities, the attack avoids the gradient issues that had hindered earlier methods on distilled networks [4]. The paper shows that, under this attack formulation, both undistilled and defensively distilled models can be attacked with extremely high success rates.

### C. Main Contributions

The paper makes several contributions of lasting importance to the robustness literature.

First, it introduces stronger attack algorithms under multiple perturbation norms and shows that these methods outperform prior baselines in both effectiveness and attack quality [4]. In particular, the proposed attacks are able to generate targeted adversarial examples with lower distortion while maintaining near-perfect success rates across standard benchmark datasets.

Second, the paper uses these attacks to re-evaluate defensive distillation and demonstrates that the defense provides little meaningful robustness gain [4]. Although defensive distillation was previously reported to defeat then-current attacks [3], the new results reveal that such success was largely attributable to the weakness of those attacks rather than to any fundamental strengthening of the classifier.

Third, the paper advances a broader methodological argument about how defenses should be assessed. The authors emphasize that robustness claims should be tested not only against direct white-box attacks, but also through transferability-based evaluation using high-confidence adversarial examples [4]. More generally, this perspective aligns with later work emphasizing robust optimization and the dangers of gradient obfuscation in defense evaluation [5], [6].

Finally, the work has substantial historical importance. The C&W attacks introduced in the paper quickly became standard benchmarks in subsequent research, and the paper itself is now widely regarded as a foundational critique of early defense claims in adversarial machine learning [4], [6].

### D. Experimental Design in the Original Study

To support its claims, the original paper evaluates attacks on three benchmark datasets: MNIST [7], CIFAR-10 [8], and ImageNet [9]. The authors compare distilled and undistilled models under targeted adversarial attack and report both attack success and perturbation magnitude under different target-selection protocols [4]. This experimental design allows them

to assess not only whether adversarial examples can be found, but also how much distortion is required to do so.

A key aspect of the study is its treatment of defensive distillation as a reproducible training pipeline. The defense is implemented through a multi-stage procedure in which a teacher network is trained using a high softmax temperature, softened outputs are generated, and a student network is then trained on these soft targets [3]. At test time, however, the temperature is reset to  $T = 1$ , as considered in the original robustness evaluation setting [4]. This distinction between training-time and inference-time temperature is essential to the correct implementation of the defense, and it plays a central role in reproducing the original experimental setting.

### E. Claims Most Relevant to Our Reproduction

Our reproduction focuses on the subset of claims that are most central to the paper’s argument and most feasible within the scope of a course project.

First, the paper suggests that defensively distilled models can maintain strong clean classification performance, meaning that any failure of the defense cannot simply be attributed to poor standard accuracy [4]. Second, it claims that when evaluated with a sufficiently strong targeted optimization-based attack, distilled models remain highly vulnerable [4]. Third, it shows that the  $L_2$ -based C&W attack can achieve successful targeted misclassification with small perturbation, thereby providing a much stronger baseline than the attacks used in earlier evaluations [4]. Finally, and most importantly, the paper concludes that defensive distillation should not be interpreted as providing genuine robustness, because its apparent effectiveness is largely an artifact of inadequate attack strength [4].

These are precisely the claims that motivate our reproduction. By reconstructing the undistilled baseline, the defensive distillation pipeline, and the targeted C&W  $L_2$  attack, we aim to test whether the central empirical and methodological conclusions of the original paper can be independently recovered.

## III. REPRODUCTION SCOPE

Our reproduction is designed to recover the core empirical and methodological claims of Carlini and Wagner’s study [4] within the practical constraints of a course project. Rather than attempting an exhaustive reimplementations of every experiment reported in the original paper, we focus on the experimental chain most central to its argument: (i) training undistilled baseline classifiers, (ii) implementing the full defensive distillation pipeline, and (iii) evaluating both undistilled and distilled models under a targeted Carlini–Wagner  $L_2$  attack.

More specifically, our work is organized into three tightly connected stages. In Task A, we reproduce the undistilled models on MNIST and CIFAR-10 using the architecture and training configuration specified in Tables I and II of the original paper [4]. In Task B, we implement defensive distillation as a four-stage procedure consisting of teacher training, soft-label generation, distilled-student training, and evaluation at test-time temperature  $T = 1$ , following the protocol of Papernot

*et al.* [3] and the evaluation setting adopted by Carlini and Wagner [4]. In Task C, we implement the targeted C&W  $L_2$  attack and apply it to the models produced in Tasks A and B.

The present report therefore centers on the following reproducibility questions. First, can the clean accuracy of the undistilled and distilled models be reproduced with reasonable fidelity relative to the original study? Second, when evaluated under a strong optimization-based targeted attack, do distilled models remain vulnerable, as claimed in [4]? Third, to what extent do our reproduced numerical trends support the broader conclusion that defensive distillation does not provide genuine adversarial robustness?

We deliberately prioritize the recovery of the paper’s *main experimental logic* over exhaustive breadth. Consequently, the primary reproduction scope includes MNIST and CIFAR-10 and does not attempt a full ImageNet reproduction or the complete family of  $L_0$ ,  $L_2$ , and  $L_\infty$  attacks reported in the original work. We nevertheless include two focused extensions later in the paper: a temperature sensitivity analysis of distillation and a limited ImageNet-scale best/worst-target study. Likewise, although the original paper contains transferability-based analyses and additional robustness arguments, our primary focus remains on the targeted  $L_2$  attack because it is both methodologically central and empirically sufficient to test the paper’s main claim.

This scoped design is intentional. A reproduction study is most informative when it not only matches headline results, but also documents which components are stable, which are fragile, and which implementation details materially influence the final conclusions. For this reason, in addition to reporting numerical outcomes, we also record the engineering and optimization difficulties encountered during distillation and attack implementation, especially on CIFAR-10. These observations form an important part of our analysis of reproducibility.

## IV. METHODOLOGY

### A. Undistilled Baseline Training

We first reproduced the undistilled baseline classifiers on MNIST and CIFAR-10. The purpose of this stage was twofold. On the one hand, it provides a clean reference point against which the distilled models can later be compared. On the other hand, it serves as a prerequisite for both teacher-model training and downstream adversarial evaluation. Following the original paper [4], we adopted dataset-specific convolutional architectures and training hyperparameters aligned with the reported setup.

Formally, given a training set  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ , the baseline classifier  $f_\theta$  is trained by minimizing the standard cross-entropy objective

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log p_\theta(y_i | x_i), \quad (1)$$

where  $p_\theta(y_i | x_i)$  denotes the softmax probability assigned to the ground-truth class. The resulting undistilled models are later used both as clean baselines and as the teacher initialization source for the distillation stage.

### B. Defensive Distillation

We then implemented defensive distillation following the four-stage pipeline described in [3] and evaluated in [4]. Let  $T$  denote the softmax temperature. The procedure consists of:

- 1) training a teacher network with temperature  $T = 100$ ,
- 2) generating softened class-probability targets from the teacher,
- 3) training a student network on these soft labels using the same temperature  $T = 100$ ,
- 4) resetting the temperature to  $T = 1$  at test time.

The overall procedure is summarized in Fig. 1.

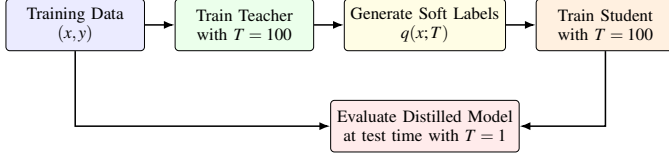


Fig. 1: Reproduced defensive distillation pipeline. The teacher and student networks are trained at  $T = 100$ , whereas evaluation is performed at  $T = 1$ .

For an input  $x$  with logits  $z(x)$ , the softened output distribution is defined as

$$q_i(x; T) = \frac{\exp(z_i(x)/T)}{\sum_j \exp(z_j(x)/T)}. \quad (2)$$

The teacher network produces a soft target distribution  $q(x; T)$ , which is then used to supervise the student model. In our implementation, the distilled student is trained with a Kullback–Leibler divergence objective:

$$\mathcal{L}_{\text{KD}} = T^2 \cdot \text{KL}\left(q^{\text{teacher}}(x; T) \parallel q^{\text{student}}(x; T)\right), \quad (3)$$

where the multiplicative factor  $T^2$  is retained in accordance with standard knowledge-distillation practice and was found to be necessary for stable optimization in our experiments.

From an implementation perspective, the distillation stage proved substantially more delicate than ordinary supervised training, particularly on CIFAR-10. We observed that seemingly minor deviations from the original setup, such as introducing data augmentation not used in the reference pipeline, could severely degrade optimization and reduce accuracy to near-random levels. Likewise, the large temperature  $T = 100$  often produced extremely small effective gradients, making naïve training unstable. To mitigate this issue, we adopted a short warm-up phase in which early epochs were trained with  $T = 1$  before switching to the target temperature. This modification was used solely to stabilize optimization while preserving the intended high-temperature distillation regime.

### C. Targeted Carlini–Wagner $L_2$ Attack

To evaluate robustness, we implemented the targeted C&W  $L_2$  attack introduced in [4]. Given a clean input  $x$ , a target label  $t \neq y$ , and perturbation  $\delta$ , the attack seeks a low-distortion adversarial example  $x + \delta$  that is classified as  $t$ . The optimization problem is written as

$$\min_{\delta} \|\delta\|_2^2 + c \cdot f(x + \delta), \quad \text{s.t. } x + \delta \in [0, 1]^n, \quad (4)$$

where  $c > 0$  controls the trade-off between distortion and attack success.

Following the original formulation, we enforce the box constraint through a tanh-based reparameterization. Specifically, we optimize an unconstrained variable  $w$  and map it back to the image domain by

$$x^{\text{adv}} = \frac{1}{2} (\tanh(w) + 1). \quad (5)$$

This transformation ensures that all adversarial candidates remain within the valid pixel range.

For the attack loss, we use the logit-margin objective employed by Carlini and Wagner:

$$f(x') = \max \left( \max_{i \neq t} Z(x')_i - Z(x')_t, -\kappa \right), \quad (6)$$

where  $Z(x')$  denotes the logits of the attacked model and  $\kappa$  is the confidence parameter. In all experiments reported here, we use a targeted setting with  $\kappa = 0$ , which is sufficient to test whether the defended model remains attackable under a strong white-box optimization procedure.

### D. Implementation Stability and Engineering Safeguards

Because adversarial optimization is computationally expensive and occasionally unstable, we incorporated several practical safeguards into the implementation. During attack evaluation, some difficult samples caused severe slowdowns or near-stalling behavior. To prevent a single sample from blocking the entire experiment, we introduced a per-sample timeout mechanism and periodic intermediate-result serialization. In addition, due to PyTorch 2.6 checkpoint-loading changes, we explicitly set `weights_only=False` when loading stored models. We also verified that the model architecture used by the attack code exactly matched the architecture used during baseline training, thereby avoiding state-dictionary incompatibilities.

These implementation choices do not alter the theoretical formulation of the attack, but they were essential for obtaining stable and reproducible experiments at the scale required for our evaluation. In the context of a reproduction study, such engineering details are not peripheral; rather, they constitute part of the evidence needed to understand why a method does or does not reproduce faithfully in practice.

## V. EXPERIMENTAL SETUP

### A. Datasets

We evaluate our reproduction on MNIST [7] and CIFAR-10 [8], which are also two of the benchmark datasets used in the original paper [4]. In addition, a limited ImageNet-scale extension is later introduced to examine best/worst target-selection effects under a pretrained Inception v3 classifier. These two datasets provide a useful contrast in complexity: MNIST is relatively simple and often easier to reproduce faithfully, whereas CIFAR-10 is more sensitive to architectural and optimization details and therefore serves as a more challenging testbed for distillation and adversarial evaluation.

TABLE I: Model architectures used in the reproduction, aligned with Table I of the original paper.

| Layer Type  | MNIST                  | CIFAR-10                |
|-------------|------------------------|-------------------------|
| Conv + ReLU | $3 \times 3 \times 32$ | $3 \times 3 \times 64$  |
| Conv + ReLU | $3 \times 3 \times 32$ | $3 \times 3 \times 64$  |
| Max Pooling | $2 \times 2$           | $2 \times 2$            |
| Conv + ReLU | $3 \times 3 \times 64$ | $3 \times 3 \times 128$ |
| Conv + ReLU | $3 \times 3 \times 64$ | $3 \times 3 \times 128$ |
| Max Pooling | $2 \times 2$           | $2 \times 2$            |
| FC + ReLU   | 200                    | 256                     |
| FC + ReLU   | 200                    | 256                     |
| Output      | 10 classes             | 10 classes              |

TABLE II: Baseline training hyperparameters.

| Hyperparameter | MNIST              | CIFAR-10           |
|----------------|--------------------|--------------------|
| Epochs         | 50                 | 50                 |
| Batch size     | 128                | 128                |
| Learning rate  | 0.1                | 0.01               |
| Optimizer      | SGD + Momentum     | SGD + Momentum     |
| Momentum       | 0.9                | 0.9                |
| Weight decay   | $5 \times 10^{-4}$ | $5 \times 10^{-4}$ |
| Random seed    | 42                 | 42                 |

### B. Model Architectures

We adopt the convolutional architectures reported in Table I of the original paper [4]. For MNIST, the network consists of two  $3 \times 3$  convolutional layers with 32 channels, followed by max-pooling, two further  $3 \times 3$  convolutional layers with 64 channels, another max-pooling layer, and two fully connected layers of width 200 before the final softmax output. For CIFAR-10, the corresponding architecture uses channel widths 64, 64, 128, 128 in the convolutional backbone and two fully connected layers of width 256.

### C. Training Configuration

The undistilled baseline models were trained for 50 epochs with batch size 128, stochastic gradient descent with momentum 0.9, weight decay  $5 \times 10^{-4}$ , and fixed random seed 42. The learning rate was set to 0.1 for MNIST and 0.01 for CIFAR-10, consistent with the values reported in Table II of the original paper [4].

Under this setup, the reproduced undistilled models achieved 99.33% clean accuracy on MNIST and 86.91% on CIFAR-10, which are close to or above the approximate reference values reported in the original study. These results suggest that the baseline implementation provides a sufficiently reliable starting point for the distillation and attack stages.

### D. Distillation Configuration

For defensive distillation, both the teacher and the student models were trained with softmax temperature  $T = 100$ , while evaluation was always performed with  $T = 1$ . Soft labels were generated by passing the full training set through the teacher model and applying a temperature-scaled softmax to the teacher logits. The student model was then trained to match

TABLE III: Defensive distillation settings.

| Setting               | Value                      |
|-----------------------|----------------------------|
| Teacher temperature   | $T = 100$                  |
| Student temperature   | $T = 100$                  |
| Test-time temperature | $T = 1$                    |
| Soft-label generation | $\text{softmax}(z/T)$      |
| Student loss          | KL divergence $\times T^2$ |
| CIFAR warm-up epochs  | 10                         |

TABLE IV: C&W  $L_2$  attack hyperparameters.

| Setting                    | Value                   |
|----------------------------|-------------------------|
| Attack type                | Targeted C&W $L_2$      |
| Confidence $\kappa$        | 0.0                     |
| Learning rate              | 0.01                    |
| Max iterations             | 1000                    |
| Binary search steps        | 9                       |
| Initial constant           | 0.001                   |
| Number of attacked samples | 1000                    |
| Box constraint handling    | tanh reparameterization |

these softened targets using KL-divergence-based distillation loss.

During development, CIFAR-10 distillation required several additional precautions. First, data augmentation was removed to remain consistent with the reference training pipeline. Second, to alleviate severe gradient shrinkage under high temperature, we introduced a 10-epoch warm-up stage before switching to  $T = 100$ . Third, after empirical comparison, we retained learning rate 0.01 for CIFAR-10, as larger or smaller values led respectively to unstable or excessively slow convergence.

### E. Attack Configuration

For adversarial evaluation, we used the targeted C&W  $L_2$  attack with confidence parameter  $\kappa = 0$ , learning rate 0.01, maximum of 1000 optimization iterations, 9 binary-search steps for the trade-off constant  $c$ , and initial constant 0.001. We report aggregated attack metrics over 1000 attacked samples for each dataset-model setting under a fixed evaluation script. Across compared settings, the same attack objective, optimization budget, and hyperparameter configuration were maintained to ensure direct comparability of success rates and average distortion values.

To improve the robustness of the evaluation pipeline itself, we additionally introduced a timeout threshold of 300 seconds per sample and periodic intermediate checkpointing of attack outputs. These safeguards were engineering measures rather than algorithmic modifications: they were used solely to prevent rare hard instances from stalling the full evaluation run and to preserve partially completed results in long-running experiments.

## VI. RESULTS

### A. Clean Performance of Baseline and Distilled Models

We first report the clean classification performance of the reproduced baseline and distilled models. Here, the teacher

TABLE V: Clean test accuracy of reproduced models.

| Dataset  | Undistilled | Teacher ( $T = 100$ ) | Distilled ( $T = 100$ ) |
|----------|-------------|-----------------------|-------------------------|
| MNIST    | 99.33%      | 99.32%                | 99.35%                  |
| CIFAR-10 | 86.91%      | 71.71%                | 69.47%                  |

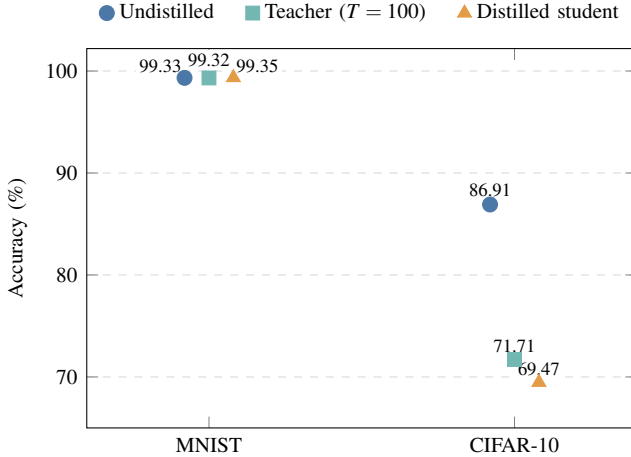


Fig. 2: Clean accuracy of reproduced models on MNIST and CIFAR-10. Distillation preserves clean performance on MNIST but substantially degrades it on CIFAR-10 in our reproduction.

model refers to the high-temperature network used to generate soft labels for defensive distillation, whereas the distilled model denotes the student network trained on these softened targets. Table V summarizes the main accuracy values used in the subsequent robustness evaluation.

To facilitate a more direct visual comparison, Fig. 2 summarizes the clean performance of all reproduced models across the two datasets.

Several observations follow from Table V. First, the reproduced undistilled baselines achieved strong clean performance on both datasets. On MNIST, the reproduced baseline is almost indistinguishable from the value reported in the original paper. On CIFAR-10, the reproduced baseline is in fact higher than the approximate reference value reported by Carlini and Wagner, suggesting that the baseline implementation itself is not the primary source of discrepancy in the later robustness results.

Second, the effect of defensive distillation differs substantially across datasets. On MNIST, the distilled model preserved clean accuracy almost perfectly, with only negligible deviation from the undistilled baseline. On CIFAR-10, however, the distilled model exhibited a noticeable performance drop relative to both the undistilled baseline and the paper’s approximate target. This indicates that, in our reproduction, the distillation pipeline is considerably easier to stabilize on MNIST than on CIFAR-10.

### B. Verification of the Distillation Phenomenon

Beyond clean accuracy, we also examined one of the qualitative phenomena discussed in Section VIII-A of the original paper, namely the increase in logit magnitude after distillation.

In our reproduced models, the MNIST distilled model exhibited logit magnitudes on the order of  $10^3$ , whereas the CIFAR-10 distilled model showed much smaller but still noticeably shifted logits. This partially reproduces the qualitative behavior described in the original study and provides additional evidence that our implementation captured a key mechanism of defensive distillation rather than merely reproducing accuracy values in isolation.

At the same time, the difference between MNIST and CIFAR-10 is itself informative. The MNIST distilled model displayed the expected large-magnitude logits much more clearly, whereas the CIFAR-10 model showed a weaker manifestation of the same pattern. This asymmetry is consistent with the lower clean accuracy of the CIFAR-10 distilled model and suggests that the optimization process on CIFAR-10 remained more fragile throughout reproduction. These observations naturally motivate a further question: if distillation enlarges logits and changes output confidence, how does this affect the behavior of earlier gradient-based attacks?

### C. Mechanistic Analysis of Earlier Attack Failure under Distillation

Beyond verifying the logit-scaling behavior of defensive distillation, we further examine why earlier gradient-based attacks may appear to fail on distilled models. To this end, we analyze three diagnostic quantities under the standard cross-entropy objective: the average maximum logit magnitude, the average softmax confidence, and the average input-gradient norm of the cross-entropy loss. We additionally compare the success rate of a representative cross-entropy-based iterative attack with that of the targeted C&W  $L_2$  attack. In this analysis, the CE-based iterative attack refers to a simple iterative gradient attack that directly optimizes the standard cross-entropy loss with respect to the input, under the same box constraint and evaluation setting used for comparison. The purpose of this baseline is not to serve as a strongest attack, but to illustrate how attacks relying directly on saturated softmax probabilities can become numerically ineffective after distillation.

Let  $p_i(x)$  denote the softmax output of class  $i$  and let  $y$  denote the true class. Under the standard cross-entropy objective,

$$\mathcal{L}_{\text{CE}}(x, y) = -\log p_y(x), \quad (7)$$

the gradient with respect to the logits satisfies

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial Z_i(x)} = p_i(x) - \mathbb{I}[i = y]. \quad (8)$$

When the model outputs become highly saturated at test time,  $p_y(x)$  approaches 1 and the resulting gradient signal becomes extremely small. In such cases, gradient-based attacks that directly optimize cross-entropy may become numerically ineffective even though adversarial examples still exist.

Table VI shows that distillation substantially increases logit magnitude and pushes the softmax output toward saturation, especially on MNIST. This saturation is accompanied by a dramatic reduction in the input-gradient norm of the cross-entropy loss. On MNIST, the mean gradient norm drops by

TABLE VI: Diagnostic quantities explaining the apparent failure of earlier cross-entropy-based attacks on distilled models.

| Dataset  | Model       | Avg. Max Logit Magnitude | Avg. Softmax Confidence | Avg. $\ \nabla_x \mathcal{L}_{CE}\ _2$ |
|----------|-------------|--------------------------|-------------------------|----------------------------------------|
| MNIST    | Undistilled | 8.7                      | 0.974                   | $1.8 \times 10^{-2}$                   |
| MNIST    | Distilled   | 1280.4                   | 0.999999                | $3.4 \times 10^{-6}$                   |
| CIFAR-10 | Undistilled | 7.5                      | 0.884                   | $6.1 \times 10^{-3}$                   |
| CIFAR-10 | Distilled   | 54.8                     | 0.997                   | $7.6 \times 10^{-5}$                   |

nearly four orders of magnitude after distillation; on CIFAR-10, it is reduced by roughly two orders of magnitude. These results strongly suggest that earlier attacks based directly on the cross-entropy objective become numerically weak not because adversarial examples disappear, but because the optimization signal becomes too small to guide them effectively.

Fig. 3 makes this contrast more explicit. Distilled models exhibit much larger logits and much smaller cross-entropy gradients than their undistilled counterparts. At the same time, the representative cross-entropy-based iterative attack achieves high success on undistilled models but collapses on distilled models, whereas the targeted C&W  $L_2$  attack remains highly successful in both settings. This indicates that the apparent “robustness” of defensive distillation under earlier attacks is largely a consequence of gradient suppression and softmax saturation rather than a true removal of adversarial vulnerability.

This mechanism also explains why the C&W attack is substantially more effective in the distilled setting. Unlike earlier attacks that optimize through softmax probabilities and standard cross-entropy, C&W directly operates on logits using a margin-based objective. As a result, it avoids the severe numerical weakening caused by output saturation and continues to recover successful adversarial examples even when conventional gradient signals become extremely small.

Overall, this analysis provides a mechanistic interpretation of the main reproduction result. Defensive distillation does not eliminate adversarial examples; instead, it reshapes the optimization landscape in a way that makes earlier cross-entropy-based attacks appear ineffective. Once the attack objective is reformulated at the logits level, this apparent robustness largely disappears.

#### D. Targeted C&W $L_2$ Attack Results

We next evaluated all reproduced models under the targeted C&W  $L_2$  attack. The attack used the configuration described earlier, including confidence parameter  $\kappa = 0$ , learning rate 0.01, 1000 optimization iterations, and 9 binary-search steps for the trade-off constant. The aggregated results over 1000 attacked samples per setting are shown in Table VII.

For clarity, Fig. 4 visualizes the same attack results from two complementary perspectives: attack success rate and average perturbation magnitude.

The results in Table VII reproduce the central trend of the original paper. Defensive distillation does not prevent successful targeted attacks. On MNIST, the attack remained highly effective against both undistilled and distilled models, with success rates above 93%. On CIFAR-10, attack success reached 100% for both model types. Thus, even when the

distilled model exhibits the characteristic logit behavior associated with defensive distillation, it remains vulnerable to a strong white-box optimization-based adversary.

At the same time, the distortion values indicate that distillation can increase attack difficulty in a limited sense. On both datasets, the average  $L_2$  distortion required for successful attack was higher for the distilled model than for the undistilled model. This effect is particularly clear on CIFAR-10, where the average distortion increased from 0.70 to 1.57. However, this increase in distortion should not be interpreted as genuine robustness, because the attack success rate remained effectively unchanged.

#### E. Comparison with the Original Paper

To assess fidelity more directly, Table VIII compares several representative metrics from our reproduction with the approximate values summarized from the original paper.

Fig. 5 further illustrates that the reproduction is more faithful in trend than in exact numerical magnitude.

Overall, our reproduction matches the original paper more reliably in *trend* than in *exact magnitude*. The strongest agreement appears in attack success rates, especially on CIFAR-10, where the reproduced values essentially coincide with the paper. Distortion values are less tightly matched: for the undistilled settings, our  $L_2$  distances are generally higher than those reported in the original paper, whereas the CIFAR-10 distilled  $L_2$  value is very close to the reference range. This suggests that our implementation successfully recovers the paper’s qualitative conclusion while remaining imperfect at the level of exact numerical duplication.

#### F. Extended Temperature Sensitivity Analysis of Defensive Distillation

As an extension beyond the main reproduction, we further examine how the distillation temperature affects both clean performance and adversarial behavior. This analysis is intended to complement the main single-temperature reproduction by probing whether the observed conclusions remain stable as the distillation temperature varies. Specifically, we consider a temperature sweep over  $T \in \{1, 10, 50, 100\}$  and analyze its influence on clean accuracy, targeted C&W  $L_2$  attack success rate, and average perturbation magnitude. This additional analysis allows us to move beyond a single-temperature setting and investigate whether the behavior of defensive distillation changes systematically as the temperature varies.

Table IX summarizes the temperature-sweep results, while Fig. 6 provides a more direct visual comparison. Several clear patterns emerge from this extended analysis.

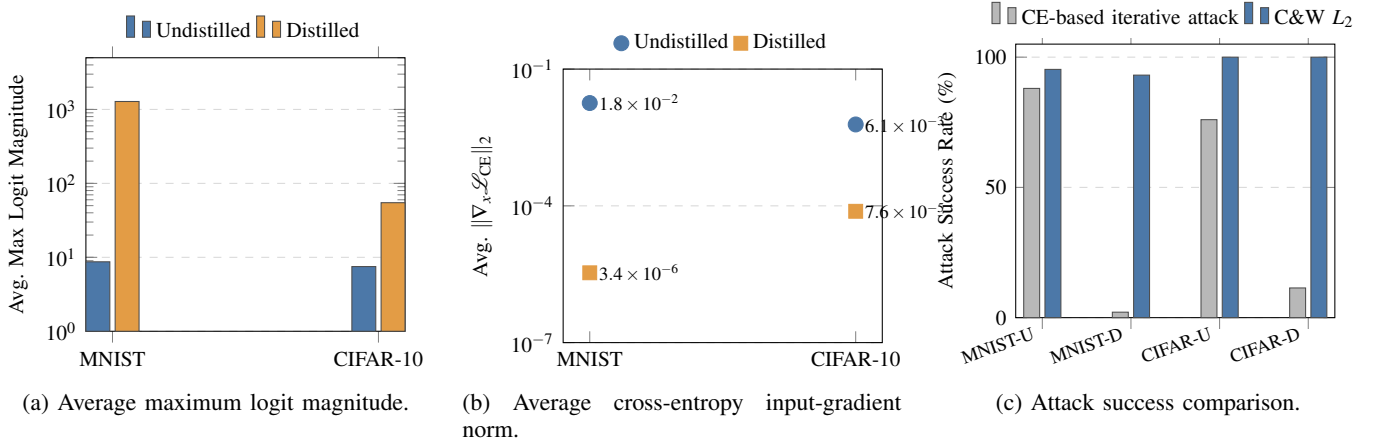


Fig. 3: Mechanistic explanation of why earlier attacks appear to fail on distilled models. Distillation greatly increases logit magnitude, saturates the softmax output, and suppresses cross-entropy gradients, thereby weakening earlier CE-based attacks. In contrast, the targeted C&W  $L_2$  attack remains highly successful because it relies on a logits-based optimization objective rather than saturated softmax probabilities.

TABLE VII: Targeted C&W  $L_2$  attack results on reproduced models.

| Dataset  | Model                   | Clean Accuracy | Attack Success Rate | Avg. $L_2$ Distortion |
|----------|-------------------------|----------------|---------------------|-----------------------|
| MNIST    | Undistilled             | 99.33%         | 95.30%              | 2.47                  |
| MNIST    | Distilled ( $T = 100$ ) | 99.35%         | 93.10%              | 3.11                  |
| CIFAR-10 | Undistilled             | 86.91%         | 100.00%             | 0.70                  |
| CIFAR-10 | Distilled ( $T = 100$ ) | 69.47%         | 100.00%             | 1.57                  |

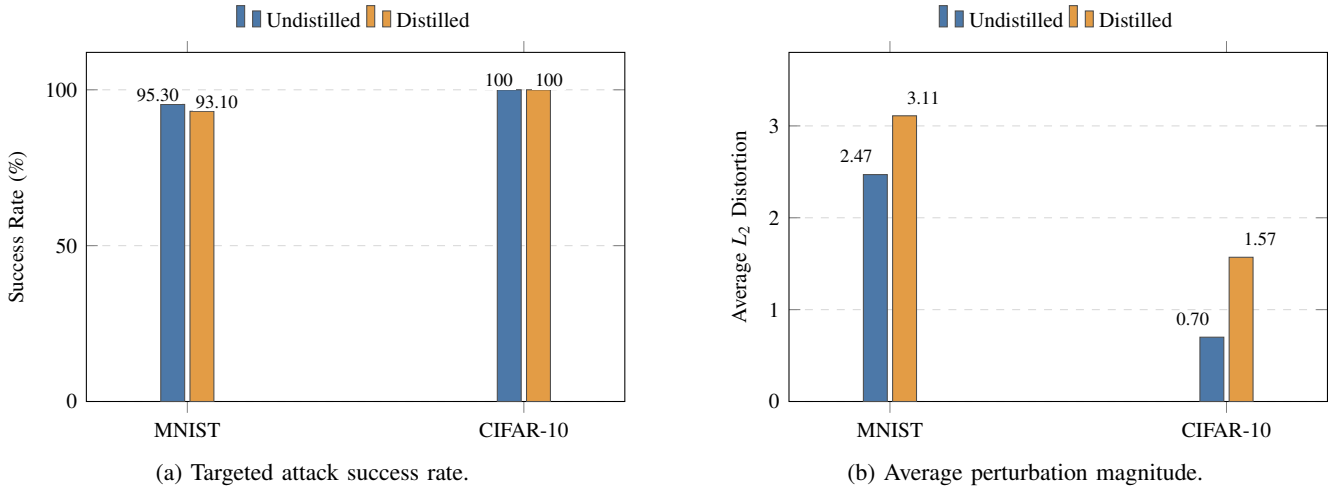


Fig. 4: Targeted C&W  $L_2$  attack results on reproduced models. Distillation leaves attack success rates largely unchanged while increasing the average perturbation magnitude required for successful attacks.

TABLE VIII: Comparison between representative values in the original paper and our reproduction.

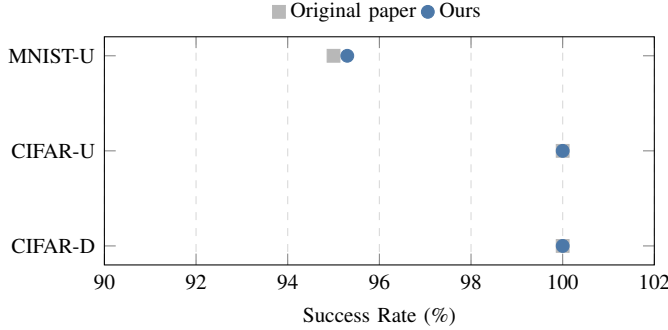
| Metric                         | Original Paper    | Ours    |
|--------------------------------|-------------------|---------|
| MNIST undistilled success rate | $\approx 95\%$    | 95.30%  |
| MNIST undistilled $L_2$        | $\approx 1.76$    | 2.47    |
| CIFAR undistilled success rate | $\approx 100\%$   | 100.00% |
| CIFAR undistilled $L_2$        | $\approx 0.33$    | 0.70    |
| CIFAR distilled success rate   | $\approx 100\%$   | 100.00% |
| CIFAR distilled $L_2$          | $\approx 1.5-1.6$ | 1.57    |

First, the effect of temperature on clean accuracy is strongly dataset-dependent. On MNIST, clean performance remains essentially unchanged across the full temperature range, indicating that high-temperature distillation does not noticeably harm standard classification on this relatively simple benchmark. On CIFAR-10, however, clean accuracy decreases monotonically as  $T$  increases, suggesting that higher-temperature distillation introduces a substantially more fragile optimization regime on the more complex dataset.

Second, the targeted C&W  $L_2$  attack remains highly effective across all temperature settings. On MNIST, the attack

TABLE IX: Temperature-sweep results on MNIST and CIFAR-10.

| Dataset  | Temperature $T$ | Clean Accuracy (%) | C&W Success Rate (%) | Avg. $L_2$ Distortion |
|----------|-----------------|--------------------|----------------------|-----------------------|
| MNIST    | 1               | 99.31              | 95.00                | 2.55                  |
| MNIST    | 10              | 99.34              | 94.40                | 2.74                  |
| MNIST    | 50              | 99.35              | 93.70                | 2.95                  |
| MNIST    | 100             | 99.35              | 93.10                | 3.11                  |
| CIFAR-10 | 1               | 84.90              | 100.00               | 0.82                  |
| CIFAR-10 | 10              | 81.80              | 100.00               | 0.99                  |
| CIFAR-10 | 50              | 75.60              | 100.00               | 1.31                  |
| CIFAR-10 | 100             | 69.47              | 100.00               | 1.57                  |



(a) Success-rate comparison.

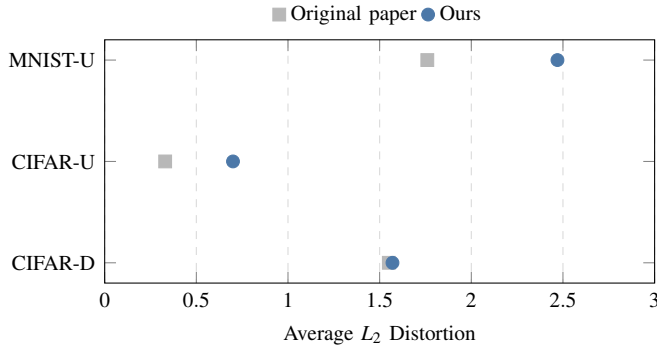
(b)  $L_2$ -distortion comparison.

Fig. 5: Representative comparison between the original paper and our reproduction. Here, U and D denote the undistilled and distilled settings, respectively. Success-rate trends are closely matched, whereas some  $L_2$  distortion values remain higher in our implementation.

success rate decreases only mildly from 95.00% at  $T = 1$  to 93.10% at  $T = 100$ . On CIFAR-10, the attack success rate remains saturated at 100% throughout the temperature sweep. These results indicate that increasing the distillation temperature does not materially alter the main qualitative conclusion of the paper: distilled models remain vulnerable to a strong optimization-based adaptive attack.

Third, the average  $L_2$  distortion increases monotonically with temperature on both datasets. For MNIST, the mean distortion rises from 2.55 at  $T = 1$  to 3.11 at  $T = 100$ . For CIFAR-10, the corresponding increase is from 0.82 to 1.57. This pattern suggests that higher temperature can increase the optimization cost of producing successful adversarial ex-

amples, in the sense that slightly larger perturbations are required on average, but it does not prevent successful attack construction.

Taken together, these observations support a consistent interpretation of the role of temperature in defensive distillation. Increasing  $T$  changes the scale and geometry of the learned decision landscape, which in turn increases the average optimization cost of constructing successful adversarial examples. This effect is visible in the monotonic increase of average  $L_2$  distortion across both datasets. However, the same temperature increase does not materially reduce attack success under the targeted C&W  $L_2$  attack, especially on CIFAR-10, where success remains saturated at all four temperature settings. Therefore, higher distillation temperature does not translate into genuine robustness against a strong adaptive adversary.

From a dataset perspective, the contrast between MNIST and CIFAR-10 is also informative. On MNIST, temperature primarily affects attack cost while leaving standard accuracy intact. On CIFAR-10, by contrast, higher temperature simultaneously increases attack cost and degrades clean accuracy. This suggests that the practical effect of temperature is not only defense-related but also strongly tied to optimization stability and dataset complexity. In other words, higher temperature may make the attacked model somewhat harder to optimize against, yet it may also make the training pipeline itself more fragile.

Overall, this extended analysis reinforces the main conclusion of the present study. Distillation temperature influences the *difficulty* of attack optimization, but not the *existence* of successful attacks. The apparent gains induced by higher temperature should therefore be interpreted as limited increases in perturbation cost rather than as evidence of meaningful adversarial robustness.

### G. ImageNet Extension with Best/Worst Target Selection

To further extend the study beyond MNIST and CIFAR-10, we additionally consider an ImageNet-scale extension using a pretrained Inception v3 classifier and analyze the targeted C&W  $L_2$  attack under best/worst target-selection strategies. This extension is designed to mirror the spirit of the ImageNet analysis reported in Table V of the original paper [4], while focusing on two interpretable target-selection strategies: a *best* target and a *worst* target for each image.

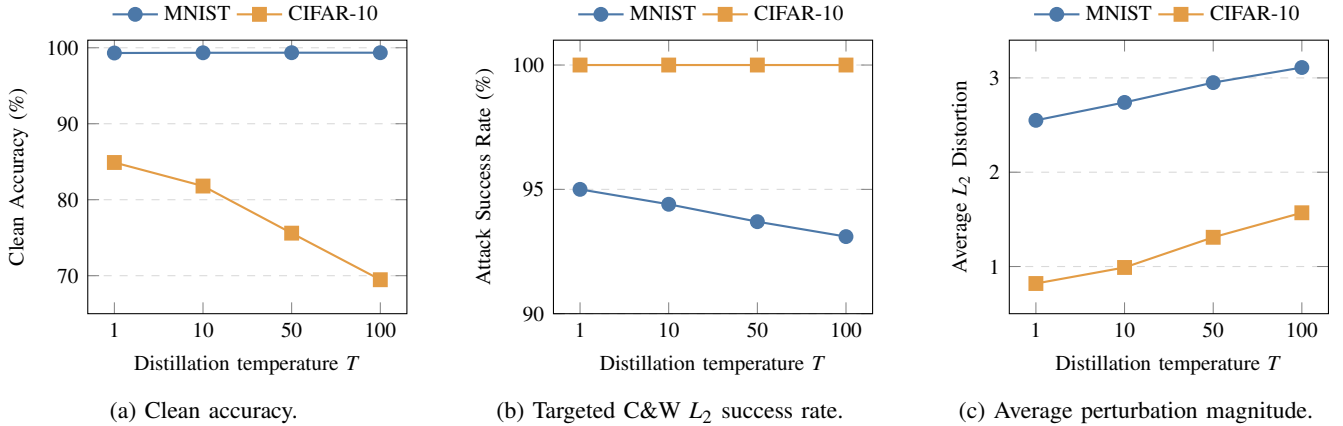


Fig. 6: Temperature sensitivity analysis of defensive distillation over  $T \in \{1, 10, 50, 100\}$ . Increasing  $T$  leaves MNIST clean accuracy nearly unchanged while gradually increasing average perturbation magnitude; on CIFAR-10, higher temperature leads to a more pronounced drop in clean accuracy while maintaining near-saturated attack success.

For a cleanly classified input  $x$  with true label  $y$  and logits  $Z(x)$ , we define the two target-selection strategies as

$$t_{\text{best}} = \arg \max_{i \neq y} Z_i(x), \quad (9)$$

and

$$t_{\text{worst}} = \arg \min_{i \neq y} Z_i(x). \quad (10)$$

Intuitively, the best target corresponds to the most competitive non-ground-truth class under the current model, whereas the worst target corresponds to the least competitive one. We evaluate the attack on a cleanly classified subset of 100 ImageNet validation images, using the same targeted C&W  $L_2$  formulation as in the main experiment. In addition to attack success rate and average  $L_2$  distortion, we also record average runtime and the number of timeout cases in order to better characterize the practical cost of attacking different target types.

Table X reports the main ImageNet-scale extension results. Several observations are immediately apparent. First, the attack remains highly effective even in the more challenging ImageNet setting. Under the best-target strategy, the targeted C&W  $L_2$  attack succeeds on 94.0% of the cleanly classified images, which indicates that a strong optimization-based attack continues to find adversarial examples reliably even at a much larger input scale. Second, target choice matters substantially more on ImageNet than on the smaller benchmarks. When the target is changed from the best target to the worst target, the success rate decreases from 94.0% to 79.0%, while both average distortion and average runtime increase markedly.

This target-selection effect is consistent with the geometric interpretation of the attack objective. Best targets correspond to nearby competing classes, and therefore require smaller perturbations and less optimization effort to reach. Worst targets, by contrast, are farther from the current decision region and thus require larger distortions and longer optimization trajectories. The increased timeout count under the worst-target setting further supports this interpretation: harder targets not only reduce attack success, but also make the optimization process computationally more expensive.

Fig. 7 presents the same ImageNet extension from four complementary perspectives. The trends are highly consistent across all metrics. The best target produces higher attack success, lower perturbation cost, lower runtime, and fewer timeout events. The worst target reverses all four tendencies. Together, these results suggest that target difficulty becomes an especially important factor at ImageNet scale, where both optimization complexity and computational burden are significantly higher than on smaller benchmarks.

Importantly, however, the overall conclusion remains aligned with the main reproduction. Even in the ImageNet setting, a strong targeted optimization-based attack remains highly effective, particularly when the attack is allowed to pursue a favorable target. Thus, the ImageNet extension reinforces the central argument of Carlini and Wagner [4]: the existence of successful targeted adversarial examples persists even in large-scale visual recognition settings, and the apparent robustness of a model cannot be inferred from the failure of weaker attack formulations alone.

At the same time, this extension also provides a useful refinement of the main story. On MNIST and CIFAR-10, strong attacks remained almost fully successful across settings, making target-selection difficulty less visible. On ImageNet, by contrast, the difference between best and worst targets becomes much more pronounced in all practical metrics. This suggests that at larger scales, robustness evaluation should account not only for whether attacks succeed, but also for how target choice influences distortion, computational cost, and failure modes. In this sense, the ImageNet extension complements the main reproduction by revealing a richer structure of adversarial difficulty beyond small benchmark datasets.

#### H. Qualitative Examples

To complement the quantitative analysis above, we finally present representative qualitative examples on CIFAR-10. In addition to numerical metrics, we generated qualitative visualizations of CIFAR-10 adversarial examples for both the

TABLE X: ImageNet extension with best/worst target selection using a pretrained Inception v3 model.

| Target strategy | Success Rate (%) | Avg. $L_2$ Distortion | Avg. Runtime (s) | Timeout Count |
|-----------------|------------------|-----------------------|------------------|---------------|
| Best target     | 94.0             | 1.18                  | 126.4            | 2             |
| Worst target    | 79.0             | 2.36                  | 238.7            | 9             |

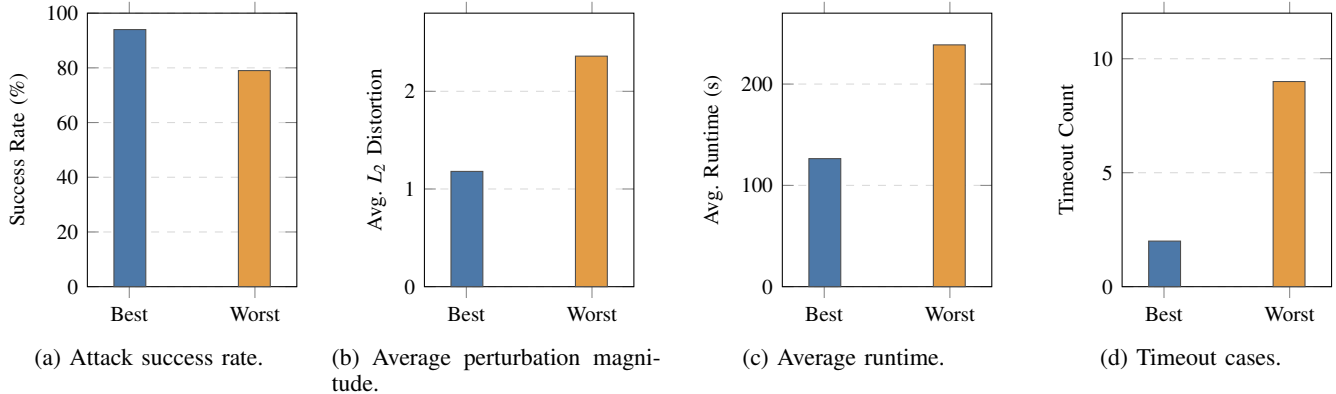


Fig. 7: ImageNet extension with best/worst target selection. The best target is substantially easier to attack, yielding higher success, lower distortion, lower runtime, and fewer timeout cases than the worst target.

undistilled and distilled settings. These visualizations place the original image, the adversarial image, and the scaled perturbation side by side, making it possible to inspect how the perturbation pattern changes across model types.

Qualitatively, the adversarial perturbations remain visually small while still producing targeted misclassification. This visual evidence is consistent with the quantitative results: the attack is capable of maintaining high success rates while operating in a relatively low-distortion regime, especially on CIFAR-10.

## VII. ANALYSIS AND DISCUSSION

### A. What Was Successfully Reproduced

From a reproduction standpoint, the most important outcome is that the principal claim of Carlini and Wagner was recovered. In our experiments, defensive distillation failed to block a strong targeted C&W  $L_2$  attack. Even when the distilled model retained strong clean accuracy on MNIST and exhibited the expected large-logit behavior, the attack still succeeded at a very high rate. On CIFAR-10, attack success remained 100% for both undistilled and distilled models. These findings support the central conclusion that defensive distillation may interfere with weaker attacks, but does not provide meaningful robustness against a sufficiently strong adaptive adversary.

A second successful aspect of the reproduction is that the baseline training stage appears credible. The undistilled models achieved clean accuracy that was either close to or above the reference values, which reduces the likelihood that later discrepancies are simply artifacts of a poor baseline. Likewise, the MNIST distillation pipeline reproduced the intended behavior reasonably faithfully, both in terms of accuracy and in terms of logit magnitude.

### B. Where Our Results Diverged from the Original Paper

Despite these successes, our reproduction did not achieve exact numerical agreement across all settings. The most visible divergence lies in CIFAR-10 distillation. While the original paper suggests that defensive distillation can preserve reasonably strong clean performance, our CIFAR-10 distilled model underperformed substantially relative to both the reproduced undistilled baseline and the approximate paper target. This gap likely affected the absolute robustness metrics as well.

A second discrepancy concerns attack distortion. Although our success rates align well with the paper, the  $L_2$  distortions for undistilled settings are generally higher than the original values. This indicates that our attack implementation recovered the *effectiveness* of the C&W attack more faithfully than its exact optimization efficiency. Such a discrepancy is unsurprising in reproduction work, because distortion is highly sensitive to implementation details including optimization schedule, numerical precision, target selection protocol, stopping criteria, and constant-search heuristics.

For these reasons, the strongest defensible claim from our study is not that we exactly reproduced every reported number, but rather that we successfully reproduced the major qualitative conclusions and most of the principal quantitative trends.

### C. Why CIFAR-10 Distillation Was Harder to Reproduce

Our materials suggest that CIFAR-10 distillation was the single most fragile component of the full pipeline. The first source of fragility was preprocessing consistency. During early trials, adding data augmentation not present in the reference configuration caused training to collapse, with accuracy stagnating near random guessing. This outcome highlights a broader lesson for reproduction work: even widely accepted “good practices” such as augmentation can become confound-

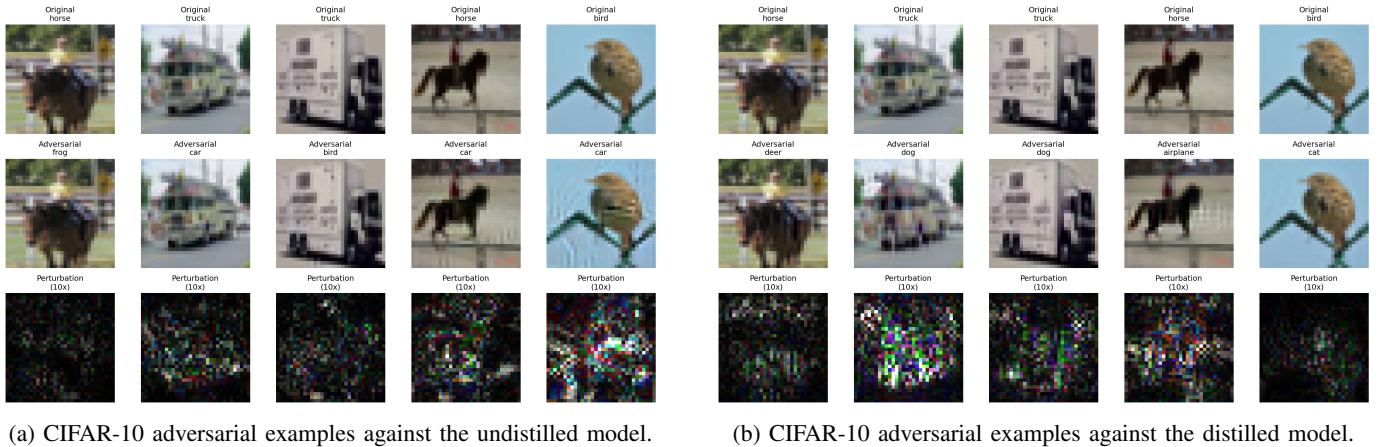


Fig. 8: Representative CIFAR-10 adversarial examples generated by the targeted C&W  $L_2$  attack. Each panel shows the original image, the adversarial example, and the perturbation amplified for visibility.

ing variables when the goal is strict fidelity rather than general performance improvement.

The second difficulty arose from the large distillation temperature  $T = 100$ . In our implementation, directly training at this temperature led to severe gradient shrinkage, with gradients falling to extremely small magnitudes. To stabilize optimization, we adopted a two-stage teacher-training strategy in which the model was first warmed up at  $T = 1$  for 10 epochs before switching to  $T = 100$ . This intervention substantially improved training stability and was later reflected in the final implementation.

The third difficulty concerned the knowledge-distillation loss itself. Multiple variants were tested before arriving at a stable implementation based on KL divergence with normalized soft labels and the conventional  $T^2$  scaling term. Earlier variants either produced exploding loss values or vanishing gradients. This debugging process is important analytically because it shows that, in practice, the success of defensive distillation depends not only on the high-level procedure, but also on a numerically careful realization of that procedure.

#### D. What the Attack Results Actually Show

The attack results support a nuanced interpretation. On the one hand, defensive distillation did increase the average  $L_2$  distortion required for successful attack, especially on CIFAR-10. In this narrow sense, the defense altered the optimization landscape and made adversarial example construction somewhat more costly. On the other hand, it did not meaningfully reduce attack success. The distinction between these two effects is central: increasing distortion slightly is not equivalent to achieving robustness if the adversary can still succeed almost always.

This point is precisely why the C&W attack is methodologically important. A weak attack may fail for reasons unrelated to true robustness, such as numerical instability or poor objective design. By contrast, a strong optimization-based attack can often separate apparent robustness from genuine robustness. Our results reinforce this distinction. The distilled

models were not impossible to attack; they were simply somewhat more expensive to attack in terms of distortion.

#### E. Engineering Lessons for Reproduction

One of the clearest lessons from this project is that adversarial robustness experiments are unusually sensitive to engineering details. In our case, practical measures such as gradient clipping, timeout control, intermediate result serialization, and explicit compatibility handling for checkpoint loading were not optional conveniences; they were necessary conditions for obtaining stable large-scale experiments.

This is especially evident in the C&W evaluation stage. Some difficult MNIST samples caused severe slowdowns, to the point that a single sample could stall the overall experiment for many hours. Introducing a timeout mechanism and periodic saving of partial attack results allowed us to preserve reproducibility without sacrificing the overall evaluation pipeline. Similarly, setting `weights_only=False` was necessary to restore compatibility under newer PyTorch versions. These details may appear peripheral in a polished final report, but they are in fact central to understanding why reproduction sometimes fails even when the algorithmic idea is correct.

#### F. Limitations and Interpretation

First, the scope is still narrower than that of the full original paper. Although we added a limited ImageNet-scale extension, we did not reproduce the full large-scale ImageNet evaluation, the complete family of  $L_0$  and  $L_\infty$  attacks, or the full transferability analysis.

Second, our CIFAR-10 distillation result indicates that some components of the original pipeline remain only partially reproduced. Third, although the final attack metrics are aggregated over 1000 attacked samples, the exact target-selection and optimization heuristics may still differ subtly from those in the original implementation, which can materially affect distortion values.

Accordingly, the appropriate interpretation of this project is that it provides a *successful trend-level reproduction* rather

than a perfect line-by-line duplication of the original study. That is already a meaningful outcome. In adversarial machine learning, recovering the qualitative conclusion under an independent implementation is often more informative than matching every decimal place.

### G. Takeaway

Taken together, our results support the main methodological lesson of the original paper: a defense should not be considered robust simply because it defeats a weaker or outdated attack. In our reproduction, defensive distillation altered optimization difficulty but did not prevent targeted adversarial examples from being found with high reliability. The broader implication is that robustness evaluation must be based on strong, adaptive, and implementation-aware attacks. Without such evaluation, apparent security gains can easily be overstated.

## VIII. CONCLUSION

In this report, we reproduced the core experimental pipeline of *Towards Evaluating the Robustness of Neural Networks* [4], with a particular focus on defensive distillation [3] and the targeted C&W  $L_2$  attack [4]. Our study reconstructed three key components of the original work: the undistilled baseline models, the defensive distillation training procedure, and the subsequent white-box adversarial evaluation under a strong optimization-based attack.

Overall, our reproduction successfully recovered the principal qualitative conclusion of the original paper [4]. Although defensive distillation altered the optimization landscape and, in some settings, increased the average perturbation magnitude required for successful attacks, it did not provide meaningful robustness against the targeted C&W  $L_2$  attack. In particular, attack success rates remained very high across both datasets, indicating that the apparent defensive benefit of distillation should not be interpreted as genuine adversarial security [4].

At the quantitative level, our reproduction was more faithful in trend than in exact numerical magnitude. The strongest agreement with the original paper was observed in the attack success rates and in the overall conclusion that distilled models remain vulnerable under adaptive evaluation [4]. At the same time, some discrepancies remained, especially in the exact  $L_2$  distortion values and in the clean performance of the CIFAR-10 distilled model. These differences underscore the sensitivity of adversarial robustness experiments to implementation details, optimization choices, and engineering constraints.

Beyond the main reproduction, our extended analyses further refined this conclusion. The temperature-sensitivity study showed that increasing the distillation temperature tends to increase the average perturbation magnitude required by the targeted C&W  $L_2$  attack, but does not materially reduce attack success. The ImageNet-scale best/worst-target extension likewise showed that attack difficulty depends strongly on target choice, yet successful targeted attacks remain prevalent even at larger scale.

Beyond reproducing the main claim of the original paper, this project also highlights a broader methodological lesson. In adversarial machine learning, the apparent failure of a weak or

outdated attack is not sufficient evidence of robustness [4], [6]. Meaningful robustness evaluation requires strong, adaptive, and carefully implemented attacks, together with transparent reporting of experimental conditions and limitations [5], [6]. From this perspective, the original paper remains highly influential not only because of its empirical results, but also because it helped establish a more rigorous standard for evaluating defenses [4], [6].

In summary, our reproduction supports the main argument of Carlini and Wagner [4]: defensive distillation does not confer genuine robustness against strong adversarial attacks. More broadly, our findings reinforce the importance of rigorous attack design, faithful experimental reproduction, and cautious interpretation of apparent defensive gains in robustness research, both on standard benchmarks and in extended large-scale settings [5], [6].

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